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Limited reflection type sensor

Abstract

A limited reflection type sensor which irradiates exit light from a light emitting unit toward an object sensing region through an exit light lens and receives, at a light receiving unit, reflected light from an object existing in the object sensing region through a reflected light lens, may include a casing including a latching portion formed on a side surface and a first opening facing the object sensing region, the casing accommodating the light emitting unit and the light receiving unit; and a lens part including a light incident/exit part formed integrally with the exit light lens and the reflected light lens, and a side surface covering part in which an end of the light incident/exit part is extended along the side surface, in which the lens part covers the first opening with the light incident/exit part and the side surface covering part is latched by the latching portion.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a sensor, and in particular, relates to a limited reflection type sensor which irradiates exit light from a light emitting unit toward an object sensing region through an exit light lens and receives, at a light receiving unit, reflected light from an object existing in the object sensing region through a reflected light lens.

BACKGROUND ART

(2) Conventionally, as a sensor for detecting whether or not an object exists at a predetermined position, there has been proposed a limited reflection type sensor which irradiates exit light from the light emitting unit to a limited sensing region of an object, and receives, at the light receiving unit, reflected light reflected by the object in the sensing region (see Patent Document 1, for example).

(3) Such limited reflection type sensor irradiates exit light from the light emitting unit to the sensing region through the exit light lens, and receives, at the light receiving unit, reflected light reflected by an object in the sensing region through the receiving light lens. When reflected light enters the light receiving unit, the light is converted into an electric signal by the light receiving unit, and hence the presence of an object in the sensing region can be sensed by detecting a voltage change occurring in the light receiving unit. Therefore, the sensing region is limited to a range in which light is irradiated from the light emitting unit and the light can reach the light receiving unit. The setting range of such sensing region can be changed by the optical design of the exit light lens and the receiving light lens.

(4) In the limited reflection type sensor of the prior art shown in Patent Document 1, each of the exit light lens and the receiving light lens is configured by a combination of an aspherical lens and a cylindrical lens. Due to this, by sensing a short distance using the cylindrical lens and sensing a long distance using the aspherical lens, it is possible to expand the long distance sensing range while maintaining the short distance sensing range.

PRIOR ART DOCUMENTS

Patent Document

(5) Patent Document 1: Japanese Unexamined Patent Publication No. 2015-81801

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

(6) In the conventional limited reflection type sensor described above, it is necessary to accommodate the exit light lens and the receiving light lens in the casing and to accurately perform focus position alignment with the light emitting unit and the light receiving unit, respectively. Therefore, a projection and a hole are integrally formed at the end and a center of the lens, and fitted into corresponding latching portions in the casing, thereby performing alignment. By assembling lenses with different focus positions, it is possible to change the sensing range of the limited reflection type sensor to one corresponding to the lens.

(7) However, when accommodating the lens in the casing, it is necessary to insert the lens into the casing in a state where the lens is held and to perform alignment to the latching portion, and hence there has been a problem that the assembly process becomes complicated. Since it is necessary to integrally form a projection and a hole for alignment on the lens, the shape of the lens becomes complicated and the difficulty level of processing becomes high. Furthermore, after aligning the lens in the casing, it is necessary to cover the casing upper portion with a lid to prevent dust and dirt from entering inside the casing. Thus, in a state where the lid is mounted, it has been difficult to check whether or not the lens corresponding to the sensing range is assembled, whether or not the lens is accurately aligned, and the like.

(8) The present invention has been devised to solve such problems, and its object is to provide a limited reflection type sensor capable of easily and accurately performing alignment at the time of mounting a lens, and capable of enhancing the dust-resistant property.

Means for Solving the Problem

(9) In order to solve the above problems, the limited reflection type sensor of the present invention is a limited reflection type sensor which irradiates exit light from a light emitting unit toward an object sensing region through an exit light lens and receives, at a light receiving unit, reflected light from an object existing in the object sensing region through a reflected light lens, the limited reflection type sensor having a casing including a latching portion formed on a side surface and a first opening facing the object sensing region, the casing accommodating the light emitting unit and the light receiving unit, and a lens part including a light incident/exit part formed integrally with the exit light lens and the reflected light lens, and a side surface covering part in which an end of the light incident/exit part is extended along the side surface, in which the lens part covers the first opening with the light incident/exit part, and the side surface covering part is latched by the latching portion.

(10) Due to this, the side surface covering part extended from the light incident/exit part is latched by the latching portion formed on the side surface of the casing, and it becomes possible to easily and accurately perform alignment at the time of mounting the lens, and possible to enhance the dust-resistant property.

(11) In one aspect of the present invention, the latching portion has a snap-fit structure in which a latching pawl is formed at the tip of an elastic piece, and the latching pawl is fitted into a latching opening formed on the side surface covering part.

(12) In one aspect of the present invention, a guide groove guiding the side surface covering part is formed on the side surface.

(13) In one aspect of the present invention, the latching portion latches the side surface covering part at a position closer to the light incident/exit part relative to half in the extension direction of the side surface covering part.

(14) In one aspect of the present invention, the first opening includes a sealing groove inside, the sealing groove being formed along the longitudinal direction of the casing, the lens part includes a sealing projection having a shape corresponding to the sealing groove formed on the first opening side, and the sealing groove and the sealing projection abut against each other.

(15) In one aspect of the present invention, the casing includes a second opening on the side opposite to the first opening, and the limited reflection type sensor further has a mounting substrate accommodated in the casing with the light emitting unit and the light receiving unit mounted thereon, and a cover latched to the casing and covering the second opening.

(16) In one aspect of the present invention, a connector for external electrical connection is mounted on the mounting substrate, and the connector is exposed from a third opening formed on another side surface of the casing.

(17) In one aspect of the present invention, the connector is exposed from a fourth opening formed on the bottom surface of the cover.

Effect of the Invention

(18) According to the present invention, it is possible to provide a limited reflection type sensor capable of performing accurate alignment at the time of mounting a lens, and capable of enhancing the dust-resistant property.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic perspective view showing a configuration of a limited reflection type

sensor **100** according to a first embodiment.

(2) FIG. 2 is an exploded perspective view schematically showing the configuration of the limited reflection type sensor **100**.

(3) FIG. 3 is a schematic perspective view for explaining assembly of a casing **10** and a substrate **30**.

(4) FIG. 4 is a schematic perspective view for explaining assembly of the casing **10** and a cover **40**.

(5) FIG. 5 is a schematic perspective view for explaining assembly of the casing **10** and a lens part **20**.

(6) FIGS. 6(a) and 6(b) are cross-sectional views of the limited reflection type sensor **100**, where FIG. 6(a) is a schematic cross-sectional view at a position A-A in FIG. 1, and FIG. 6(b) is a schematic cross-sectional view at a position B-B in FIG. 1.

(7) FIGS. 7(a) and 7(b) are views schematically showing a fixing structure of a mounting substrate **31** according to a second embodiment, where FIG. 7(a) shows an example using a press-fitting pin **19a** and a swaging pin **19b**, and FIG. 7(b) shows an example in which a bonding agent **19c** is applied to a rear surface of the mounting substrate **31**.

(8) FIG. 8 is a schematic perspective view showing a configuration of a limited reflection type sensor **200** according to the second embodiment.

(9) FIG. 9 is an exploded perspective view schematically showing the configuration of the limited reflection type sensor **200**.

(10) FIG. 10 is a schematic perspective view for explaining assembly of a casing **50** and a substrate **70**.

(11) FIG. 11 is a schematic perspective view for explaining assembly of the casing **50** and a cover **80**.

(12) FIG. 12 is a schematic perspective view for explaining assembly of the casing **50** and a lens part **60**.

(13) FIGS. 13(a) and 13(b) are cross-sectional views of the limited reflection type sensor **200**, where FIG. 13(a) is a schematic cross-sectional view at a position A-A in FIG. 8, and FIG. 13(b) is a schematic cross-sectional view at a position B-B in FIG. 8.

MODE FOR CARRYING OUT THE INVENTION

First Embodiment

(14) The first embodiment of the present invention will be described below with reference to the drawings. FIG. 1 is a schematic perspective view showing the configuration of the limited reflection type sensor **100** according to the first embodiment. FIG. 2 is an exploded perspective view schematically showing the configuration of the limited reflection type sensor **100**. As shown in FIGS. 1 and 2, the limited reflection type sensor **100** includes the casing **10**, the lens part **20**, the substrate **30**, and the cover **40**.

(15) The casing **10** is a casing forming an outer shape of the limited reflection type sensor **100** and holding each component, and FIG. 1 shows a box-shaped example of a substantially rectangular parallelepiped. The casing **10** is made of a material that does not transmit light having a wavelength longer than that of ultraviolet light, and for example, a black colored resin can be used. The casing **10** includes a casing body **11**, an attachment portion **12**, a latching portion **13**, a partition plate **14**, and an opening **15**. The shape of the casing **10** is not limited to the rectangular parallelepiped shape shown in FIG. 1.

(16) In the casing body **11**, a latching opening **11a** is formed in the vicinity of the bottom surface of the side surface, a bottom opening **11b** is formed on the bottom surface, and a side surface opening **11c** is formed in the longitudinal side surface each as an opening. The attachment portion **12** and the latching portion **13** are formed on the side surface of the short side, the opening **15** is formed on the upper surface, and the partition plate **14** is provided in the opening **15**.

(17) The latching opening **11a** is an opening formed on the side surface of the casing body **11** and is formed at a position corresponding to a latching pawl **41a** described later. Although FIGS. 1 and

2 show an example in which the latching opening **11a** is provided in the longitudinal direction of the casing **10**, the latching opening **11a** may be provided on the bottom surface in the short side direction. The bottom surface opening **11b** is an opening formed on the lower surface of the casing body **11**, and corresponds to the second opening in the present invention. The side surface opening **11c** is an opening formed on the side surface of the casing body **11**, and corresponds to the third opening in the present invention. The side surface opening **11c** is formed at a position and in a shape corresponding to a connector **34** and an opening covering part **42** described later.

(18) The attachment portion **12** is a portion formed integrally with the casing body **11** on the short side surface of the casing **10**, and is a portion for aligning and fixing the limited reflection type sensor **100** to another member. FIGS. **1** and **2** show an example in which the attachment portion **12** is a projection having a substantially cubic shape, and an attachment hole **12a**, which is a hole penetrating in the vertical direction and the depth direction, is formed. The attachment hole **12a** is a hole into which a fixing member such as a screw or a bolt is inserted, and a screw groove may be formed inside thereof as necessary. The shape of the attachment portion **12** is not limited to that shown in FIGS. **1** and **2**, and, for example, a flat plate formed to protrude substantially horizontally from the side surface may be used.

(19) The latching portion **13** is a portion formed on the side surface of the short side of the casing body **11**, and for latching the lens part **20**. Although the specific structure of the latching portion **13** is not limited, it is possible to use a snap-fit structure in which a part of the side surface is cut out as an elastic piece and a latching pawl **13a** is formed at the tip thereof as shown in FIGS. **1** and **2**, for example. In the latching portion **13**, a guide groove **13b** having the same width and length as those of a side surface covering part **24** described later is formed in the short side surface of the casing body **11**.

(20) The partition plate **14** is a substantially flat plate portion erected at the substantially center of the opening **15**. The partition plate **14** divides the light emitting side and the light receiving side, and its upper end abuts against the lower surface of a light incident/exit part **21** of the lens part **20** described later. Since the light receiving side and the light emitting side are divided by the partition plate **14**, it can prevent light from directly reaching from a light emitting unit **32** to a light receiving unit **33**.

(21) The opening **15** is an opening formed on the upper surface of the casing body **11**, and corresponds to the first opening in the present invention. A lens abutting portion **15a** and a sealing groove **15b** are formed in the opening **15**. The lens abutting portion **15a** is a portion forming an upper end of a longitudinal side surface of the casing body **11**. The sealing groove **15b** is a step or groove formed inside the lens abutting portion **15a** and is formed to a position lower than the upper end of the partition plate **14**. The shape of the sealing groove **15b** is formed to have a width, a length, and a depth corresponding to a sealing projection **25** of the lens part **20** described later. While in FIG. **2**, the sealing groove **15b** is extended to the outside relative to the elastic piece of the latching portion **13**, it may have a shape extending to the inside of the elastic piece in order to prevent erroneous insertion of the side surface covering part **24**.

(22) The lens part **20** is an optical member that is made of a translucent material such as resin, that transmits light exit from the light emitting unit **32** and takes it out to the outside with a predetermined light distribution, and that transmits light incident from the outside and causes it to enter the light receiving unit **33**. As shown in FIGS. **1** and **2**, the lens part **20** includes the light incident/exit part **21**, an exit light lens **22**, a reflected light lens **23**, the side surface covering part **24**, and the sealing projection **25**.

(23) The light incident/exit part **21** is a substantially flat plate portion forming the upper portion of the lens part **20**, and the exit light lens **22** and the reflected light lens **23** are integrally formed on the lower surface side thereof. The upper surface of the light incident/exit part **21** is a substantially flat surface, and is arranged opposite to the object sensing region. Accordingly, the limited reflection type sensor **100** performs light emission to the object sensing region and light reception

from the object sensing region via the light incident/exit part **21**. Since the lens part **20** has the light incident/exit part **21**, the lens part **20** does not protrude to the outside of the limited reflection type sensor **100**, and it is possible to save space and suppress adhesion of dirt. The outer peripheral vicinity of the light incident/exit part **21** is formed to be approximately equal to the outer diameter of the lens abutting portion **15a** of the casing body **11**, and abuts against the lens abutting portion **15a** to align and fix the lens part **20** to the casing **10**.

(24) The exit light lens **22** is a lens for irradiating the object sensing region with light exit from the light emitting unit **32**. The reflected light lens **23** is a lens for causing light reflected by the sensing target in the object sensing region to enter the light receiving unit **33**. The exit light lens **22** and the reflected light lens **23** are formed at positions separated from each other by a separation groove **26**, and the lens part **20** is molded integrally with the light incident/exit part **21**. By integrally molding the exit light lens **22** and the reflected light lens **23** as the lens part **20**, with simply aligning the lens part **20** to the casing **10**, it is possible to align the exit light lens **22** and the reflected light lens **23** to the light emitting unit **32** and the light receiving unit **33**, respectively, and it is possible to simplify the assembly process and improve alignment accuracy.

(25) The side surface covering part **24** is a flat plate-like portion in which both ends of the light incident/exit part **21** are extended downward in the figure, and is molded integrally with the light incident/exit part **21**. The shape of the side surface covering part **24** is formed to have the same width and length as those of the guide groove **13b**. A latching opening **24a** is formed in the side surface covering part **24** at a position corresponding to the latching portion **13**. When the lens part **20** is attached to the casing **10**, the side surface covering part **24** is fitted into the guide groove **13b**, and the latching portion **13** is fitted into the latching opening **24a**, thereby performing alignment and fixing.

(26) The sealing projection **25** is a linear projection provided on the rear surface side in the vicinity of the long side of the light incident/exit part **21**, and is formed in a position and shape corresponding to the sealing groove **15b**. As shown in FIG. 2, the longitudinal end of the sealing projection **25** is positioned outside relative to the inner surface of the side surface covering part **24** and positioned inside relative to the outer surface.

(27) The separation groove **26** is a groove separating between the exit light lens **22** and the reflected light lens **23** and reaching the rear surface of the light incident/exit part **21**, and aligns and holds the lens part **20** by inserting the partition plate **14**. Of the exit light lens **22** and the reflected light lens **23**, the surface exposed by the separation groove **26** has a tapered shape, thereby facilitating insertion and alignment of the partition plate **14**.

(28) The substrate **30** is a member in which the light emitting unit **32**, the light receiving unit **33**, and the connector **34** are mounted on one surface of the mounting substrate **31**. The mounting substrate **31** is a substrate on which a wiring layer (not shown) is formed, and the light emitting unit **32**, the light receiving unit **33**, and the connector **34** are mounted on the wiring layer. The material forming the mounting substrate **31** is not limited, and it is possible to use known materials such as a printed circuit board, a metal substrate, and a composite substrate of resin and metal. An aligning hole **31a**, which is a through hole, is formed on the mounting substrate **31** at a predetermined position.

(29) The light emitting unit **32** is a member to which power and a signal are transmitted from the outside of the limited reflection type sensor **100** via the connector **34** and the wiring layer, and which emits light with a predetermined wavelength. Although the specific configuration of the light emitting unit **32** is not limited, a light emitting diode (LED) or the like is used. The wavelength of the light emitted by the light emitting unit **32** only needs to be included in the absorption band of the light receiving unit **33**, and for example, infrared light is used. The light emitting unit **32** may include an optical member such as a lens, and irradiates, with light emitting from an LED chip, the light having a desired light distribution characteristic.

(30) The light receiving unit **33** is a member that absorbs light of a predetermined wavelength,

converts the light into an electric signal, and sends a detection signal to the outside of the limited reflection type sensor **100** via the connector **34** and the wiring layer. Although the specific configuration of the light receiving unit **33** is not limited, for example, a phototransistor can be used. The light receiving unit **33** includes a wavelength of light irradiated by the light emitting unit **32** in an absorption band. The light receiving unit **33** may include an optical member such as a lens, and efficiently collects incident light to a phototransistor for detection.

(31) The connector **34** is a member to which a harness or the like is connected from the outside of the limited reflection type sensor **100** and an output and a signal from the outside are transmitted. The terminal of the connector **34** is electrically connected to the wiring layer on the mounting substrate **31**. In the examples shown in FIGS. **1** and **2**, the connector **34** is mounted in a direction parallel to the mounting surface of the mounting substrate **31**, and is exposed to the outside of the casing **10** from the side surface opening **11c**.

(32) The cover **40** is a member that covers the bottom surface opening **11b** of the casing **10** and holds the substrate **30**. The cover **40** has a cover wall **41** erected along the four sides of the bottom surface outer periphery, and the outer periphery of the cover wall **41** is substantially identical to the inner periphery of the bottom surface opening **11b**. The latching pawl **41a** is formed protruding on the side surface of the cover wall **41**, and substrate supports **41b** and **41c** are formed protruding upward at the upper end of the cover wall **41**. At a position corresponding to the side surface opening **11c**, the opening covering part **42** is formed to a position higher than the substrate supports **41b** and **41c**.

(33) In the limited reflection type sensor **100** shown in FIGS. **1** and **2**, the power and the signal supplied to the connector **34** from the outside are transmitted to the light emitting unit **32**, and the light emitting unit **32** emits light. The light from the light emitting unit **32** is incident on the exit light lens **22**, and the light is irradiated to the outside through the light incident/exit part **21** with a light distribution corresponding to the curved surface of the exit light lens **22**. When an object is present in the object sensing region for sensing the object, the light irradiated from the light emitting unit **32** is reflected in the limited reflection type sensor **100** direction, enters the reflected light lens **23** via the light incident/exit part **21**, and enters the light receiving unit **33**. The light receiving unit **33** outputs a voltage corresponding to the intensity of the received light, and the voltage value is transmitted to the outside via the connector **34** as a signal for sensing an object.

(34) Next, assembly of the limited reflection type sensor **100** will be described with reference to FIGS. **3** to **5**. FIG. **3** is a schematic perspective view for explaining assembly of the casing **10** and the substrate **30**. As shown in FIG. **3**, a division wall **11d** is formed in the casing **10**. The division wall **11d** forms a connector accommodation **16**, a light emitting accommodation **17**, and a light receiving accommodation **18** inside the casing **10**. A projection downward from the division wall **11d** is formed as a substrate aligning part **19**.

(35) The connector accommodation **16** is a space formed inside the casing body **11** from the side surface opening **11c**, and accommodating the connector **34**. The connector accommodation **16** is divided from the light emitting accommodation **17** and the light receiving accommodation **18** by the division wall **11d**, and is also divided from the space accommodating the exit light lens **22** and the reflected light lens **23** by the division wall **11d** on the opening **15** side.

(36) The light emitting accommodation **17** is a space provided in communication with the opening **15**, and is further in communication with the bottom surface opening **11b** of the casing body **11**. The light emitting unit **32** is arranged at a predetermined position in the light emitting accommodation **17**. The light receiving accommodation **18** is a space provided in communication with the opening **15**, and is further in communication with the bottom surface opening **11b** of the casing body **11**. The light receiving unit **33** is arranged at a predetermined position in the light receiving accommodation **18**. The substrate aligning part **19** is a projection formed at a position corresponding to the aligning hole **31a** formed on the mounting substrate **31**.

(37) As shown in FIG. **3**, the substrate aligning part **19** and the aligning hole **31a** are aligned, the

connector accommodation **16** and the connector **34** are aligned, and the substrate **30** is inserted into the bottom surface opening **11b**. Due to this, the light emitting unit **32** is aligned at a predetermined position of the light emitting accommodation **17**, and the light receiving unit **33** is aligned at a predetermined position of the light receiving accommodation **18**.

(38) In the embodiment, the side surface opening **11c** is formed on the casing **10**, and the connector **34** is accommodated in the connector accommodation **16** and exposed from the side surface opening **11c**. As described above, since the connector accommodation **16** is divided from the light emitting accommodation **17** and the light receiving accommodation **18** by the wall surface, even if the connector **34** is exposed to the outside, it is possible to suppress entry of dust and dirt from the outside into the light emitting accommodation **17** and the light receiving accommodation **18**, and it is possible to enhance the dust-resistant property. With only alignment of the aligning hole **31a** and the substrate aligning part **19**, it is possible to easily and accurately align the light emitting unit **32** and the light receiving unit **33** at predetermined positions.

(39) FIG. **4** is a schematic perspective view for explaining assembly of the casing **10** and the cover **40**. As shown in FIG. **4**, a substantially flat plate bottom surface **43** is formed at the bottom of the cover **40**, and the cover wall **41** is provided along the outer edge of the bottom surface **43**. When the cover **40** is inserted into the bottom surface opening **11b** of the casing **10**, the outer periphery of the cover wall **41** is fitted into the inner wall of the casing body **11**, and the latching pawl **41a** is fitted into the latching opening **11a**, whereby the cover **40** is aligned and fixed. At this time, the substrate supports **41b** and **41c** formed at the upper end of the cover wall **41** support the rear surface side of the mounting substrate **31**, and the opening covering part **42** supports the lower surface side of the connector **34**. Due to this, the substrate **30** aligned inside the casing **10** is sandwiched between the cover **40** and the casing **10**, and is aligned and fixed.

(40) The opening covering part **42** is formed in a size covering a region of the side surface opening **11c** other than the region where the connector **34** is exposed, and the side surface opening **11c** can be covered with the opening covering part **42** at the same time the bottom surface opening **11b** is covered with the cover **40**. Due to this, the substrate **30** can be easily fixed with the cover **40**, and it is possible to suppress entry of dust and dirt from the outside and it is possible to enhance dust-resistant property.

(41) FIG. **5** is a schematic perspective view for explaining assembly of the casing **10** and the lens part **20**. As shown in FIG. **5**, the side surface covering part **24** of the lens part **20** is aligned to the latching portion **13** of the casing **10**, and the opening **15** is covered with the light incident/exit part **21** of the lens part **20**. At this time, at the same time, the partition plate **14** is inserted into the separation groove **26** and abuts against the rear surface of the light incident/exit part **21**, and the sealing projection **25** is inserted into the sealing groove **15b**. The side surface covering part **24** is fitted into the guide groove **13b**, and the latching pawl **13a** is fitted into the latching opening **24a**, whereby the lens part **20** is aligned and fixed to the casing **10**.

(42) FIG. **6** is a cross-sectional view of the limited reflection type sensor **100**, where FIG. **6(a)** is a schematic cross-sectional view at the position A-A in FIG. **1**, and FIG. **6(b)** is a schematic cross-sectional view at the position B-B in FIG. **1**. In the limited reflection type sensor **100** of the embodiment, the lens part **20** is mounted to the opening **15** in the casing **10**, the substrate **30** is accommodated inside the casing **10**, the connector **34** is exposed from the side surface opening **11c**, and the cover **40** is mounted to the bottom surface opening **11b** of the casing **10**.

(43) The assembly state shown in FIGS. **6(a)** and **6(b)** will be briefly described. The partition plate **14** is inserted into the separation groove **26**, and its upper end abuts against the rear surface of the light incident/exit part **21**. The upper end of the lens abutting portion **15a** abuts against the rear surface of the light incident/exit part **21**. The sealing projection **25** is fitted into the sealing groove **15b**. The side surface covering part **24** is guided and fitted into the guide groove **13b**. The latching pawl **13a** is fitted into the latching opening **24a**.

(44) The connector **34** is accommodated in the connector accommodation **16** and exposed from the

side surface opening **11c**. The substrate aligning part **19** is inserted into the aligning hole **31a**. The light emitting unit **32** is accommodated in the light emitting accommodation **17**. The light receiving unit **33** is accommodated in the light receiving accommodation **18**. The upper surface of the mounting substrate **31** abuts against the division wall **11d**.

(45) The latching pawl **41a** is fitted into the latching opening **11a**. The upper ends of the substrate supports **41b** and **41c** abut against the rear surface of the mounting substrate **31**. The upper end of the opening covering part **42** abuts against the lower surface of the connector **34**.

(46) Therefore, the upward movement of the lens part **20** is regulated by the latching pawl **13a** and the latching opening **24a**, and the downward movement of the lens part **20** is regulated by the upper end of the partition plate **14**, the lens abutting portion **15a**, and the rear surface of the light incident/exit part **21**, and hence the vertical movement is regulated and the lens part **20** is aligned. The movement of the lens part **20** to the depth direction in the figure is regulated by the fitting of the guide groove **13b** and the side surface covering part **24** and the abutting of the sealing groove **15b** and the sealing projection **25**, whereby the lens part **20** is aligned. Furthermore, the movement of the lens part **20** to the lateral direction in the figure is regulated by the insertion of the partition plate **14** into the separation groove **26** and the sandwiching of the casing **10** by the two side surface covering parts **24**, whereby the lens part **20** is aligned.

(47) These alignments are individually effective because of each movement regulation, and the effect can be further enhanced by combining a plurality of components. Accordingly, with simply mounting the lens part **20** to the casing **10**, the movement of the lens part **20** in the three-dimensional direction is regulated with respect to the casing **10**. Hence, the exit light lens **22** and the reflected light lens **23**, which are molded integrally with the light incident/exit part **21**, are easily and accurately aligned with respect to the light emitting unit **32** and the light receiving unit **33** accommodated inside the casing **10**.

(48) By providing the latching pawl **13a** on the side surface of the casing body **11**, the latching portion **13** of the casing **10** elastically deforms when the lens part **20** is mounted, and the lens part **20** does not elastically deform. This allows the risk of defocus and damage due to deformation of the lens part **20** to be reduced, and the lens part **20** to be easily mounted.

(49) In the examples shown in FIGS. **1** to **5**, the latching opening **24a** is formed at a position close to the light incident/exit part **21** which is upper side with respect to the half of the side surface covering part **24**. As a result, the elastic piece on which the latching pawl **13a** is formed can be lengthened to facilitate elastic deformation when the lens portion **20** is mounted, and the side surface covering part **24** can cover the entire elastic piece to eliminate a gap. Therefore, it is possible to easily attach the lens part **20** and improve the dust-resistant property.

(50) The upward movement of the mounting substrate **31** is regulated because the upper surface of the mounting substrate **31** abuts against the division wall **11d**, and the downward movement of the mounting substrate **31** is regulated because the substrate supports **41b** and **41c** abut against the rear surface of the mounting substrate **31**, and hence the vertical movement is regulated and the mounting substrate **31** is aligned. The substrate aligning part **19** is inserted into the aligning hole **31a** of the mounting substrate **31**, whereby the horizontal movement is regulated and the mounting substrate **31** is aligned.

(51) These alignments are individually effective because of each movement regulation, and the effect can be further enhanced by combining a plurality of components. Accordingly, with simply inserting the substrate **30** into the casing **10** and mounting the cover **40**, the movement of the substrate **30** in the three-dimensional direction is regulated with respect to the casing **10**. Hence, the light emitting unit **32** and the light receiving unit **33** mounted on the mounting substrate **31** are easily and accurately aligned in the light emitting accommodation **17** and the light receiving accommodation **18**, respectively. The opening covering part **42** is extended to the vicinity of the lower surface of the connector **34**, and hence the side surface opening **11c** is covered with the connector **34** and the opening covering part **42**, whereby it is possible to suppress entry of dust and

dirt to the inside and it is possible to enhance dust-resistant property.

(52) As described above, the limited reflection type sensor **100** of the embodiment can perform accurate alignment when the lens part **20** is mounted on the casing **10**, and can enhance the dust-resistant property. Due to this, assembling the lens part **20** having different focus position will facilitate change of the sensing range of the limited reflection type sensor **100** to one corresponding to the lens part **20**.

(53) The side surface covering part **24** extends from the light incident/exit part **21** and is integrally formed, and constitutes the outermost portion of the lens part **20**. Accordingly, it is possible to easily confirm, from the outside of the limited reflection type sensor **100**, whether or not the side surface covering part **24** is fitted into the guide groove **13b** and whether or not the latching pawl **13a** is fitted into the latching opening **24a**. Due to this, it is possible to easily confirm, after the lens part **20** is mounted, whether or not the exit light lens **22** and the reflected light lens **23** are accurately aligned.

Second Embodiment

(54) Next, the second embodiment of the present invention will be described with reference to the drawings. The contents overlapping those of the first embodiment are omitted. FIGS. **7(a)** and **7(b)** are views schematically showing a fixing structure of a mounting substrate **31** according to the second embodiment, where FIG. **7(a)** shows an example using a press-fitting pin **19a** and a swaging pin **19b**, and FIG. **7(b)** shows an example in which a bonding agent **19c** is applied to the rear surface of the mounting substrate **31**.

(55) In the example shown in FIG. **7(a)**, a plurality of aligning holes **31a** are formed on the mounting substrate **31**, and the press-fitting pin **19a** and the swaging pin **19b** formed in the division wall **11d** are inserted into the aligning holes **31a**. The press-fitting pin **19a** is a pin having a slightly tapered shape, and its tip is smaller in diameter than the aligning hole **31a**, and its base is larger in diameter than the aligning hole **31a**. When the press-fitting pin **19a** is aligned and inserted into the aligning hole **31a**, if the mounting substrate **31** is strongly pressed, the press-fitting pin **19a** made of resin is press-fitted with slight deformation. Due to this, alignment and fixing are simultaneously performed by press-fitting the press-fitting pin **19a** into the aligning hole **31a**.

(56) The swaging pin **19b** is a pin having a shape with a diameter smaller than the aligning hole **31a**. After the swaging pin **19b** is aligned and inserted into the aligning hole **31a**, the head of the swaging pin **19b** is crushed and deformed. Due to this, alignment and fixing are simultaneously performed by swaging the swaging pin **19b** into the aligning hole **31a**.

(57) In the example shown in FIG. **7(b)**, after the mounting substrate **31** is inserted into a bottom surface opening **11b**, the bonding agent **19c** is applied to the entire rear surface side of the mounting substrate **31**. The bonding agent **19c** fills and seals a gap between the mounting substrate **31** and the inner surface of a casing body **11**, and fixes the mounting substrate **31** inside the casing body **11**.

(58) As a fixing structure of the mounting substrate **31**, for example, a latching pawl may be formed in the casing body **11**, and the latching pawl may be engaged with the mounting substrate **31**. Furthermore, the above-described sandwiching by the cover **40**, the press-fitting pin **19a**, the swaging pin **19b**, the application of the bonding agent **19c**, and the latching by the latching pawl may be appropriately combined.

Third Embodiment

(59) Next, a third embodiment of the present invention will be described with reference to the drawings. The contents overlapping those of the first embodiment are omitted. FIG. **8** is a schematic perspective view showing the configuration of the limited reflection type sensor **200** according to the second embodiment. FIG. **9** is an exploded perspective view schematically showing the configuration of the limited reflection type sensor **200**. As shown in FIGS. **8** and **9**, the limited reflection type sensor **200** includes a casing **50**, a lens part **60**, a substrate **70**, and a cover **80**.

(60) The casing **50** includes a casing body **51**, an attachment portion **52**, a latching portion **53**, a partition plate **54**, and an opening **55**. In the casing body **51**, a latching opening **51a** is formed in the vicinity of the bottom surface of the side surface, a bottom opening **51b** is formed on the bottom surface, and a side surface opening **51c** is formed in the longitudinal side surface each as an opening. The latching portion **53** uses a snap-fit structure in which a part of the side surface is cut out as an elastic piece and a latching pawl **53a** is formed at the tip thereof. In the latching portion **53**, a guide groove **53b** having the same width and length as those of a side surface covering part **64** is formed in the short side surface of the casing body **51**. A lens abutting portion **55a** and a sealing groove **55b** are formed in the opening **55**.

(61) The lens part **60** includes the light incident/exit part **61**, an exit light lens **62**, a reflected light lens **63**, the side surface covering part **64**, and the sealing projection **65**. The exit light lens **62** and the reflected light lens **63** are formed at positions separated from each other by a separation groove **66**, and the lens part **60** is molded integrally with the light incident/exit part **61**. A latching opening **64a** is formed in the side surface covering part **64** at a position corresponding to the latching portion **53**.

(62) The substrate **70** is a member in which a light emitting unit **72** and a light receiving unit **73** are mounted on one surface of a mounting substrate **71**, and a connector **74** is mounted on the other surface thereof. An aligning hole **71a**, which is a through hole, is formed on the mounting substrate **71** at a predetermined position.

(63) The cover **80** has a cover wall **81**, a latching pawl **81a**, a cover wall upper end **82a**, a substrate abutting portion **82b**, a bottom surface **83**, and a connector opening **84**. The cover wall upper end **82a** is an upper end of a region of the cover wall **81** that is formed low. The substrate abutting portion **82b** is an upper end of a region of the cover wall **81** that is formed high, and is formed so as to surround the outer periphery of the connector **74**. The connector opening **84** is an opening formed on the bottom surface **83** at a position corresponding to the connector **74**, and is formed inside the substrate abutting portion **82b**, thereby corresponding to the fourth opening in the present invention. In the examples shown in FIGS. **8** and **9**, the connector **74** is mounted downward from the rear surface of the mounting substrate **71**, and is exposed from the connector opening **84** to the outside of the casing **50**.

(64) Next, assembly of the limited reflection type sensor **200** will be described with reference to FIGS. **10** to **12**. FIG. **10** is a schematic perspective view for explaining assembly of the casing **50** and the substrate **70**. As shown in FIG. **10**, the partition plate **54** is formed inside the casing **50**. The partition plate **54** forms a light emitting accommodation **57** and a light receiving accommodation **58** inside the casing **50**. On the partition plate **54**, a downward projection is formed as a substrate aligning part **59**. The substrate aligning part **59** is a projection formed at a position corresponding to the aligning hole **71a** formed on the mounting substrate **71**.

(65) As shown in FIG. **10**, the substrate aligning part **59** and the aligning hole **71a** are aligned, and the substrate **70** is inserted into the bottom surface opening **51b**. Due to this, the light emitting unit **72** is aligned at a predetermined position of the light emitting accommodation **57**, and the light receiving unit **73** is aligned at a predetermined position of the light receiving accommodation **58**.

(66) FIG. **11** is a schematic perspective view for explaining assembly of the casing **50** and the cover **80**. As shown in FIG. **11**, a substantially flat plate bottom surface **83** is formed at the bottom of the cover **80**, and the cover wall **81** is provided along the outer edge of the bottom surface **83**. When the cover **80** is inserted into the bottom surface opening **51b** of the casing **50**, the outer periphery of the cover wall **81** is fitted into the inner wall of the casing body **51**, and the latching pawl **81a** is fitted into the latching opening **51a**, whereby the cover **80** is aligned and fixed. At this time, the substrate abutting portion **82b** abuts against the rear surface side of the mounting substrate **71**. Due to this, the substrate **70** aligned inside the casing **50** is sandwiched between the cover **80** and the casing **50**, and is aligned and fixed.

(67) FIG. **12** is a schematic perspective view for explaining assembly of the casing **50** and the lens

part **60**. As shown in FIG. **12**, the side surface covering part **64** of the lens part **60** is aligned to the latching portion **53** of the casing **50**, and the opening **55** is covered with the light incident/exit part **61** of the lens part **60**. At this time, at the same time, the partition plate **54** is inserted into the separation groove **66** and abuts against the rear surface of the light incident/exit part **61**, and the sealing projection **65** is inserted into the sealing groove **55b**. The side surface covering part **64** is fitted into the guide groove **53b**, and the latching pawl **53a** is fitted into the latching opening **64a**, whereby the lens part **60** is aligned and fixed to the casing **50**.

(68) FIGS. **13(a)** and **13(b)** are cross-sectional views of the limited reflection type sensor **200**, where FIG. **13(a)** is a schematic cross-sectional view at the position A-A in FIG. **8**, and FIG. **13(b)** is a schematic cross-sectional view at the position B-B in FIG. **8**. In the limited reflection type sensor **200** of the embodiment, the lens part **60** is mounted to the opening **55** in the casing **50**, the substrate **70** is accommodated inside the casing **50**, and the cover **80** is mounted to the bottom surface opening **51b** of the casing **50**. The connector **74** is exposed downward from the connector opening **84**.

(69) The limited reflection type sensor **200** of the embodiment is different from the first embodiment in that the connector **74** is mounted on the rear surface side of the mounting substrate **71** and is exposed downward in the figure from the connector opening **84** provided on the cover **80**. Since the connector **74** is not arranged between the light emitting unit **72** and the light receiving unit **73**, the area of the mounting substrate **71** is reduced. That eliminates the necessity for providing a space where the connector **74** is accommodated between the light emitting accommodation **57** and the light receiving accommodation **58**, and the casing **50** is also downsized. Due to the downsizing of the casing body **51**, the attachment portion **52** is provided below the casing body **51** in a protruding manner relative to the cover **80**.

(70) As shown in FIGS. **13(a)** and **13(b)**, with simply mounting the lens part **60** to the casing **50**, the movement of the lens part **60** in the three-dimensional direction is regulated with respect to the casing **50**. Hence, the exit light lens **62** and the reflected light lens **63**, which are molded integrally with the light incident/exit part **61**, are easily and accurately aligned with respect to the light emitting unit **72** and the light receiving unit **73** accommodated inside the casing **50**.

(71) The upward movement of the mounting substrate **71** is regulated because the upper surface of the mounting substrate **71** abuts against the partition plate **54**, and the downward movement of the mounting substrate **71** is regulated because the substrate abutting portion **82b** abuts against the rear surface of the mounting substrate **71**, and hence the vertical movement is regulated and the mounting substrate **71** is aligned. Due to this, with simply inserting the substrate **70** into the casing **50** and mounting the cover **80**, the movement of the substrate **70** with respect to the casing **50** is regulated. Hence, the light emitting unit **72** and the light receiving unit **73** mounted on the mounting substrate **71** are easily and accurately aligned in the light emitting accommodation **57** and the light receiving accommodation **58**, respectively.

(72) In the limited reflection type sensor **200** of the embodiment, the substrate abutting portion **82b** surrounds the circumference of the connector **74** and abuts against the rear surface side of the mounting substrate **71**. Due to this, the substrate abutting portion **82b** covers the circumference of the connector **74**, whereby it is possible to suppress entry of dust and dirt to the inside and it is possible to enhance dust-resistant property.

(73) As described above, also the limited reflection type sensor **200** of the embodiment can perform accurate alignment when the lens part **60** is mounted on the casing **50**, and can enhance the dust-resistant property.

(74) It is to be noted that the embodiments disclosed herein are exemplary in all respects and are not a basis for limited construe. Accordingly, the technical scope of the present invention should not be construed only by the above-described embodiments, but should be defined based on the claims. The meaning equivalent to the scope of claims and all modifications within the scope are included.

DESCRIPTION OF SYMBOLS

(75) **10**, **50** casing **20**, **60** lens part **30**, **70** substrate **40**, **80** cover **100**, **200** limited reflection type sensor **11**, **51** casing body **11a**, **51a** latching opening **11b**, **51b** bottom surface opening **11c**, **51c** side surface opening **11d** division wall **12**, **52** attachment portion **12a** attachment hole **13**, **53** latching portion **13a**, **53a** latching pawl **13b**, **53b** guide groove **14**, **54** partition plate **15**, **55** opening **15a**, **55a** lens abutting portion **15b**, **55b** sealing groove **16** connector accommodation **17**, **57** light emitting accommodation **18**, **58** light receiving accommodation **19**, **59** substrate aligning part **19a** press-fitting pin **19b** swaging pin **19c** bonding agent **21**, **61** light incident/exit part **22**, **62** exit light lens **23**, **63** reflected light lens **24**, **64** side surface covering part **24a**, **64a** latching opening **25**, **65** sealing projection **26**, **66** separation groove **31**, **71** mounting substrate **31a**, **71a** aligning hole **32**, **72** light emitting unit **33**, **73** light receiving unit **34**, **74** connector **41**, **81** cover wall **41a**, **81a** latching pawl **41b** substrate support **42** opening covering part **43**, **83** bottom surface **82a** cover wall upper end **82b** substrate abutting portion **84** connector opening

Claims

1. A limited reflection type sensor which irradiates exit light from a light emitting unit toward an object sensing region through an exit light lens and receives, at a light receiving unit, reflected light from an object existing in the object sensing region through a reflected light lens, the limited reflection type sensor comprising: a casing comprising a latching portion formed on a side surface and a first opening facing the object sensing region, the casing accommodating the light emitting unit and the light receiving unit; and a lens part comprising a light incident/exit part formed integrally with the exit light lens and the reflected light lens, and a side surface covering part in which an end of the light incident/exit part is extended along the side surface, wherein the lens part covers the first opening with the light incident/exit part, and the side surface covering part is latched by the latching portion, the side surface covering part extends from the end of the light incident/exit part, is formed as a single piece, and forms the outermost part of the lens part, and the casing comprises a guide groove that guides the side covering portion, the guide groove is arranged on the side surface of the casing, and the guide groove has approximately the same width and length as the side covering portion.
2. The limited reflection type sensor according to claim 1, wherein the latching portion has a snap-fit structure in which a latching pawl is formed at a tip of an elastic piece, and the latching pawl is fitted into a latching opening formed on the side surface covering part.
3. The limited reflection type sensor according to claim 1, wherein the latching portion latches the side surface covering part at a position closer to the light incident/exit part relative to half in an extension direction of the side surface covering part.
4. The limited reflection type sensor according to claim 1, wherein the first opening comprises a sealing groove inside, the sealing groove being formed along a longitudinal direction of the casing, the lens part comprises a sealing projection having a shape corresponding to the sealing groove formed on the first opening side, and the sealing groove and the sealing projection abut against each other.
5. The limited reflection type sensor according to claim 1, wherein the casing comprises a second opening on a side opposite to the first opening, and the limited reflection type sensor further comprises: a mounting substrate accommodated in the casing with the light emitting unit and the light receiving unit mounted thereon; and a cover latched to the casing and covering the second opening.
6. The limited reflection type sensor according to claim 5, wherein a connector for external electrical connection is mounted on the mounting substrate, and the connector is exposed from a third opening formed on another side surface of the casing.
7. The limited reflection type sensor according to claim 5, wherein a connector for external

electrical connection is mounted on the mounting substrate, and the connector is exposed from a fourth opening formed on a bottom surface of the cover.

8. The limited reflection type sensor according to claim 2, wherein a guide groove guiding the side surface covering part is formed on the side surface.

9. The limited reflection type sensor according to claim 2, wherein the latching portion latches the side surface covering part at a position closer to the light incident/exit part relative to half in an extension direction of the side surface covering part.

10. The limited reflection type sensor according to claim 2, wherein the first opening comprises a sealing groove inside, the sealing groove being formed along a longitudinal direction of the casing, the lens part comprises a sealing projection having a shape corresponding to the sealing groove formed on the first opening side, and the sealing groove and the sealing projection abut against each other.

11. The limited reflection type sensor according to claim 2, wherein the casing comprises a second opening on a side opposite to the first opening, and the limited reflection type sensor further comprises: a mounting substrate accommodated in the casing with the light emitting unit and the light receiving unit mounted thereon; and a cover latched to the casing and covering the second opening.

12. The limited reflection type sensor according to claim 11, wherein a connector for external electrical connection is mounted on the mounting substrate, and the connector is exposed from a third opening formed on another side surface of the casing.

13. The limited reflection type sensor according to claim 11, wherein a connector for external electrical connection is mounted on the mounting substrate, and the connector is exposed from a fourth opening formed on a bottom surface of the cover.

14. The limited reflection type sensor according to claim 1, wherein the latching portion latches the side surface covering part at a position closer to the light incident/exit part relative to half in an extension direction of the side surface covering part.

15. The limited reflection type sensor according to claim 1, wherein the first opening comprises a sealing groove inside, the sealing groove being formed along a longitudinal direction of the casing, the lens part comprises a sealing projection having a shape corresponding to the sealing groove formed on the first opening side, and the sealing groove and the sealing projection abut against each other.

16. The limited reflection type sensor according to claim 1, wherein the casing comprises a second opening on a side opposite to the first opening, and the limited reflection type sensor further comprises: a mounting substrate accommodated in the casing with the light emitting unit and the light receiving unit mounted thereon; and a cover latched to the casing and covering the second opening.

17. The limited reflection type sensor according to claim 16, wherein a connector for external electrical connection is mounted on the mounting substrate, and the connector is exposed from a third opening formed on another side surface of the casing.

18. The limited reflection type sensor according to claim 16, wherein a connector for external electrical connection is mounted on the mounting substrate, and the connector is exposed from a fourth opening formed on a bottom surface of the cover.

19. The limited reflection type sensor according to claim 1, wherein the light incident/exit part is a substantially flat portion that forms the upper portion of the lens portion.
